Hamburg University of Technology
Institute of Mathematics
Chair of Numerical Mathematics

Prof. Dr. Sabine Le Borne
M. Sc. Jonas Grams

Hierarchical Algorithms
Winter semester 2024/2025
Exercise sheet 10, Solution hints
Discussed on January 15th, 2025

Homework:

Exercise 1: Let I, J, K be index sets. For $L \in \{I, J, K\}$ let T_L be a cluster tree for the index set L. Let the block cluster trees $T_{I\times J}$ and $T_{J\times K}$ be constructed using the cluster trees T_I, T_J, T_K and admissibility conditions (compare Equation (5.2) in the lecture notes). Let T be a tree and let $\ell \in \mathbb{N}_0$. Then, we define

$$T^{(\ell)} := \{t \in T : \text{level}(t) = \ell\}.$$

We define the product tree $T_{I\times J\times K}$ corresponding to the block cluster trees $T_{I\times J}$ and $T_{J\times K}$ inductively by

- $T^{(0)}_{I\times J\times K} := \{I \times K\}$, i.e. $I \times K$ is the root of $T_{I\times J\times K}$.
- For $\ell \in \mathbb{N}_0$, the set of successors of $r \times t \in T^{(\ell)}_{I\times J\times K}$ is defined by

$$\begin{aligned}\mathrm{S}(r \times t) := \{r' \times t' \in \mathrm{S}(r) \times \mathrm{S}(t) : \exists s \in T^{(\ell)}_J : \exists s' \in \mathrm{S}(s) :\\ r' \times s' \in \mathrm{S}(r \times s) \text{ und } s' \times t' \in \mathrm{S}(s \times t)\}.\end{aligned}$$

  We hence obtain

$$T^{(\ell+1)}_{I\times J\times K} = \cup_{r\times t \in T^{(\ell)}_{I\times J\times K}} \mathrm{S}(r \times t).$$

a) Construct the product tree $T^2_{I\times I\times I}$ for $T^2_{I\times I}$ and $T^2_{I\times I}$ from Exercise 3.

b) Show that $\quad \text{depth}(T_{I\times J\times K}) \leq \min\{\text{depth}(T_{I\times J}), \text{depth}(T_{J\times K})\}$.

c) Let the block cluster trees $T_{I\times J}$ and $T_{J\times K}$ be C_{sp}-sparse. Prove that the product tree $T_{I\times J\times K}$ is C_{sp}-sparse as well with a C_{sp}-constant that is not larger than the product of the C_{sp}-constants of $T_{I\times J}$ and $T_{J\times K}$.

Hint: The product tree $T_{I\times J\times K}$ is a block cluster tree for the index set $I \times K$, and in the definition of the product tree it holds that

$$\mathrm{S}(r) \subset T^{(\ell+1)}_I,\ \mathrm{S}(t) \subset T^{(\ell+1)}_K,\ \mathrm{S}(s) \subset T^{(\ell+1)}_J,\ \mathrm{S}(r \times s) \subset T^{(\ell+1)}_{I\times J},\ \mathrm{S}(s \times t) \subset T^{(\ell+1)}_{J\times K}$$

for all $r \times t \in T^{(\ell)}_{I\times J\times K}$ and $s \in T^{(\ell)}_J$.
Solution hints:

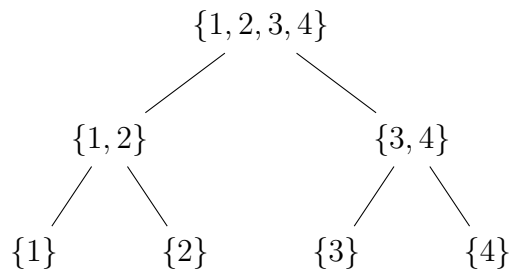


Figure 1: Cluster tree T_I

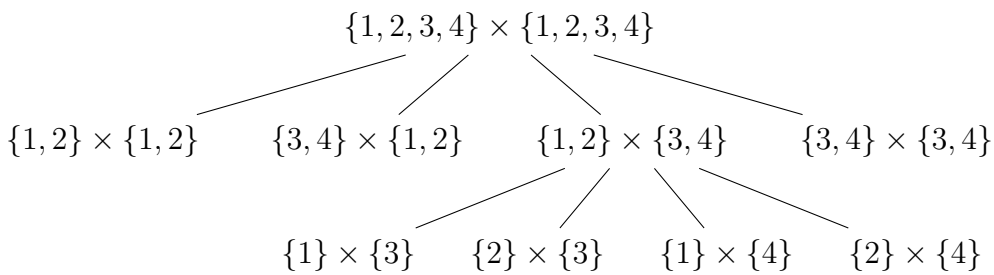


Figure 2: Block cluster tree $T_{I\times I}$

a) W.l.o.g. let $I = \{1,2,3,4\}$, let the cluster tree T_I be as depicted in Figure 1 and let the block cluster tree $T^2_{I\times I}$ be as depicted in Figure 2. Then, the product tree $T^2_{I\times I\times I}$ for the block cluster trees $T^2_{I\times I}$ and $T^2_{I\times I}$ is given be the block cluster tree shown in Figure 3. This product tree furthermore fits to the induced block partitioning of the product $A_2 \cdot B_2$ determined in Exercise 1.

b) We set $\ell := \min\{\operatorname{depth}(T_{I\times J}), \operatorname{depth}(T_{J\times K})\}$. Then, it follows from the definition of $\mathrm{S}(r\times t)$ that $\mathrm{S}(r\times t) = \emptyset$ for all $r\times t \in T^{(\ell)}_{I\times J\times K}$ since $\mathrm{S}(r\times s) = \emptyset$ or $\mathrm{S}(s\times t) = \emptyset$ for all $s \in T^{(\ell)}_J$. We hence obtain $\operatorname{depth}(T_{I\times J\times K}) \leq \min\{\operatorname{depth}(T_{I\times J}), \operatorname{depth}(T_{J\times K})\}$.

c) For $L \in \{I\times J, J\times K, I\times J\times K\}$, let C^L_{sp} be the C_{sp}-constant for the block cluster tree T_L. For $r \in T_I$,

$$\operatorname{row}(r, T_{I\times J\times K}) = \{t \in T_K : r\times t \in T_{I\times J\times K}\}$$

applies. For $t \in \operatorname{row}(r, T_{I\times J\times K})$, $t \in T_K$ and $r\times t \in T_{I\times J\times K}$ holds. With the definition of the product tree $T_{I\times J\times K}$, we find that $t \in \operatorname{row}(r, T_{I\times J\times K})$ holds if and only if a $s \in T_J$ exists with $r\times s \in T_{I\times J}$ and $s\times t \in T_{J\times K}$. The following holds:

$$\begin{aligned}
&s \in \operatorname{row}(r, T_{I\times J}) \text{ and } \#\operatorname{row}(r, T_{I\times J}) \leq C^{T_{I\times J}}_{sp},\\
&s \in \operatorname{col}(t, T_{J\times K}) \text{ and } \#\operatorname{col}(t, T_{J\times K}) \leq C^{T_{J\times K}}_{sp} \text{ as well as}\\
&\#\operatorname{row}(s, T_{J\times K}) \leq C^{T_{J\times K}}_{sp}.
\end{aligned}$$

Overall, it follows from the definition of the product tree $T_{I\times J\times K}$ that

$$\#\operatorname{row}(r, T_{I\times J\times K}) \leq \min\{C^{T_{I\times J}}_{sp}, C^{T_{J\times K}}_{sp}\} \cdot C^{T_{J\times K}}_{sp}$$

holds since there exist at most $\min\{C^{T_{I\times J}}_{sp}, C^{T_{J\times K}}_{sp}\} \cdot C^{T_{J\times K}}_{sp}$ different clusters $s\times t \in T_{J\times K}$ that satisfy $s \in T_J$ and $r\times s \in T_{I\times J}$. The number $\min\{C^{T_{I\times J}}_{sp}, C^{T_{J\times K}}_{sp}\}$ is the maximum

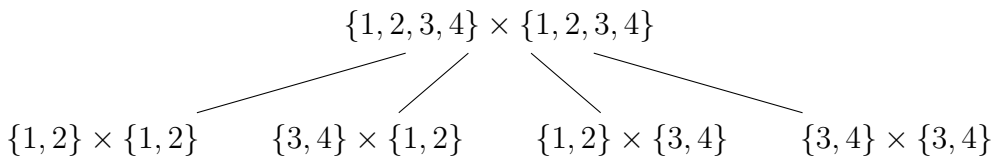


Figure 3: Product tree $T^2_{I\times I\times I}$ for the block cluster trees $T^2_{I\times I}$ and $T^2_{I\times I}$

number of different clusters s and the parameter $C_{sp}^{T_{J\times K}}$ is the maximum number of different clusters t for a fixed cluster s.

For $t \in T_K$, we obtain $\#\operatorname{col}(t, T_{I\times J\times K}) \leq \min\{C_{sp}^{T_{I\times J}}, C_{sp}^{T_{J\times K}}\} \cdot C_{sp}^{T_{I\times J}}$ analogously. For all $r \in T_I$ and for all $t \in T_K$, we hence find

$$\begin{aligned}\max\{\#\operatorname{row}(r, T_{I\times J\times K}), \#\operatorname{col}(t, T_{I\times J\times K})\} &\leq \min\{C_{sp}^{T_{I\times J}}, C_{sp}^{T_{J\times K}}\} \cdot \max\{C_{sp}^{T_{I\times J}}, C_{sp}^{T_{J\times K}}\} \\ &= C_{sp}^{T_{I\times J}} \cdot C_{sp}^{T_{J\times K}}.\end{aligned}$$

Overall, we hence find $C_{sp}^{T_{I\times J\times K}} \leq C_{sp}^{T_{I\times J}} \cdot C_{sp}^{T_{J\times K}}$.

Exercise 2: Let A be invertible and let $X^{(0)}$ be an initial value of the Newton-iteration with the error bound $\|A\|\ \|X^{(0)} - A^{-1}\| =: q < 1$ where $\|\cdot\|$ is a submultiplicative matrix norm. Prove the quadratic convergence

$$\|X^{(m)} - A^{-1}\| \leq \|A^{-1}\|\ q^{2^m}.$$

Let $A, X^{(0)}$ be symmetric positive definite with $A^{-1} > X^{(0)} > 0$. Show that the Newton method converges for all such initial values (i.e. also without error bound q for the initial value) and that $X^{(m)} > 0$ holds for all $m \geq 0$.
Hint: For $F_m := I - A^{\frac{1}{2}} X^{(m)} A^{\frac{1}{2}} > 0$ show that $F_{m+1} = F_m^2$.
Solution hints:

For $m \in \mathbb{N}_0$, we define $E_m := X^{(m)} - A^{-1}$. For all $m \in \mathbb{N}_0$, we hence find

$$\begin{aligned}E_{m+1} &= X^{(m+1)} - A^{-1} = 2X^{(m)} - X^{(m)}AX^{(m)} - A^{-1} = X^{(m)} - X^{(m)}AX^{(m)} - A^{-1} + X^{(m)} \\ &= (X^{(m)} - A^{-1})(I - AX^{(m)}) = (X^{(m)} - A^{-1})A(A^{-1} - X^{(m)}) = -E_m A E_m\end{aligned}$$

and

$$\begin{aligned}\|X^{(m+1)} - A^{-1}\| = \|E_{m+1}\| = \|-E_m A E_m\| &\leq \|A\| \cdot \|E_m\|^2 \leq \ldots \leq \|E_0\|^{2^{m+1}} \prod_{i=0}^{m} \|A\|^{2^i} \\ &= \|E_0\|^{2^{m+1}} \cdot \|A\|^{\sum_{i=0}^{m} 2^i} = \|E_0\|^{2^{m+1}} \cdot \|A\|^{2^{m+1}-1} \\ &= \|X^{(0)} - A^{-1}\|^{2^{m+1}} \cdot \|A\|^{2^{m+1}-1} \leq \left(\frac{q}{\|A\|}\right)^{2^{m+1}} \cdot \|A\|^{2^{m+1}-1} \\ &= q^{2^{m+1}} \cdot \|A\|^{-1} \leq \|A^{-1}\| \cdot q^{2^{m+1}}.\end{aligned}$$

Here, we used that

$$1 = \|I\| = \|AA^{-1}\| \leq \|A\| \cdot \|A^{-1}\|$$

holds for a submultiplicative matrix norm.

Further,

$$\begin{aligned}F_{m+1} &= I - A^{\frac{1}{2}}X^{(m+1)}A^{\frac{1}{2}} = I - A^{\frac{1}{2}}(2X^{(m)} - X^{(m)}AX^{(m)})A^{\frac{1}{2}}\\ &= I - 2A^{\frac{1}{2}}X^{(m)}A^{\frac{1}{2}} + A^{\frac{1}{2}}X^{(m)}AX^{(m)}A^{\frac{1}{2}} = I^2 - 2IA^{\frac{1}{2}}X^{(m)}A^{\frac{1}{2}} + (A^{\frac{1}{2}}X^{(m)}A^{\frac{1}{2}})^2\\ &= (I - A^{\frac{1}{2}}X^{(m)}A^{\frac{1}{2}})^2 = F_m^2\end{aligned}$$

holds for all $m \in \mathbb{N}_0$. Hence, $F_{m+1} = F_m^2 = \ldots = F_0^{2^{m+1}}$. With $A^{-1} > X^{(0)} > 0$, we obtain $A^{-\frac{1}{2}}A^{-\frac{1}{2}} > X^{(0)} > 0$. Hence, $I > A^{\frac{1}{2}}X^{(0)}A^{\frac{1}{2}} > 0$ holds, i.e. we have $F_0 = I - A^{\frac{1}{2}}X^{(0)}A^{\frac{1}{2}} > 0$. Altogether, we find that $F_m > 0$ for all $m \in \mathbb{N}_0$ since the square of a positive definite matrix is a positive definite matrix.
With induction, we find that

$$\begin{aligned}F_m = I - A^{\frac{1}{2}}X^{(m)}A^{\frac{1}{2}} > 0 &\Leftrightarrow A^{-1} - X^{(m)} > 0 \Leftrightarrow I - AX^{(m)} > 0\\ &\Leftrightarrow (X^{(m)})^{-\frac{1}{2}} - A(X^{(m)})^{\frac{1}{2}} > 0 \Leftrightarrow I - (X^{(m)})^{\frac{1}{2}}A(X^{(m)})^{\frac{1}{2}} > 0,\end{aligned}$$

$$X^{(m+1)} = X^{(m)} + (X^{(m)} - X^{(m)}AX^{(m)}) = X^{(m)} + (X^{(m)})^{\frac{1}{2}}(I - (X^{(m)})^{\frac{1}{2}}A(X^{(m)})^{\frac{1}{2}})(X^{(m)})^{\frac{1}{2}}$$

and $X^{(m+1)} > X^{(m)} > 0$ holds for all $m \in \mathbb{N}_0$. Here, we used that the matrices $A^{\frac{1}{2}}$, ABA, and $A + B$ are positive definite and that $A + B > A$ is satisfied for positive definite matrices A, B. Hence, $A^{-1} > X^{(m+1)} > X^{(m)} > \ldots > X^{(0)} > 0$ holds for all $m \in \mathbb{N}$. In particular, $X^{(m)} > 0$ holds for all $m \in \mathbb{N}_0$.
For all $m \in \mathbb{N}_0$, we hence find

$$\begin{aligned}\|X^{(m)} - A^{-1}\| = \|A^{-1} - X^{(m)}\| &= \|A^{-\frac{1}{2}}(I - A^{\frac{1}{2}}X^{(m)}A^{\frac{1}{2}})A^{-\frac{1}{2}}\| = \|A^{-\frac{1}{2}}F_mA^{-\frac{1}{2}}\|\\ &\leq \|A^{-\frac{1}{2}}\| \cdot \|F_m\| \cdot \|A^{-\frac{1}{2}}\| = \|A^{-\frac{1}{2}}\|^2 \cdot \|F_0^{2^m}\|.\end{aligned}$$

Let $\lambda \in \operatorname{spec}(F_0)$ and $\mu \in \operatorname{spec}(A^{\frac{1}{2}}X^{(0)}A^{\frac{1}{2}})$. Since $F_0 > 0$ and $A^{\frac{1}{2}}X^{(0)}A^{\frac{1}{2}} > 0$, $\lambda > 0$ and $\mu > 0$ hold. With $F_0 = I - A^{\frac{1}{2}}X^{(0)}A^{\frac{1}{2}} > 0$, we find that $\lambda = 1 - \mu > 0$. We hence find that $\lambda \in (0, 1)$ and thus $\operatorname{spec}(F_0) \subset (0, 1)$. Thus, $\lim_{m\to\infty} F_0^m = 0$ holds and $\lim_{m\to\infty} F_0^{2^m} = 0$ is satisfied, i.e. the Newton method converges.

Exercises (in class):

Exercise 3: Let $k \in \mathbb{N}$ and let I be an index set. Let T_I be a cluster tree corresponding to the index set I in which each node has two or zero successors. Let $T^j_{I\times I}$ be a block cluster tree constructed from the cluster tree T_I and an admissibility condition adm_j (see Equation (5.2) in the lecture notes). Following the sections 2.2, 3.6, and 6.3 in the lecture notes there hold the following rules:

$$\mathcal{H} \cdot \mathcal{R} = \mathcal{R},\ \mathcal{H} \cdot \mathcal{H} = \mathcal{H},\ \mathcal{R} \cdot \mathcal{R} = \mathcal{R},\ \mathcal{H} + \mathcal{R} = \mathcal{H},\ \mathcal{H} + \mathcal{H} = \mathcal{H},\ \mathcal{R} + \mathcal{R} = \mathcal{R} \text{ etc.} \tag{1}$$

For $A_j, B_j \in \mathcal{H}(T^j_{I\times I}, k)$ with block partitionings as in Figure 4, we aim to compute the $\mathcal{H}$-matrix-matrix product $A_j \cdot B_j$ where the induced block partitioning of $A_j \cdot B_j$ results from the block partitionings of A_j and B_j using the rules from (1). Determine the induced block partitioning of the $\mathcal{H}$-matrix-matrix products $A_j \cdot B_j$ for all $j \in \{1, 2, 3, 4\}$.

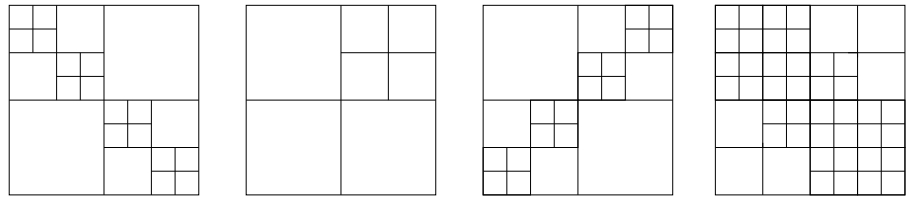


Figure 4: Block partitionings $A_j, B_j \in \mathcal{H}(T^j_{I\times I}, k)$ for $j \in \{1,2,3,4\}$.

Remark: In general, a block partitioning for the target format of an $\mathcal{H}$-matrix-matrix multiplication is given. If the block partitioning of the target format is coarser than the induced block partitioning, we can perform a (formatted) agglomeration subsequently.

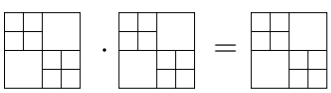
Solution hints:

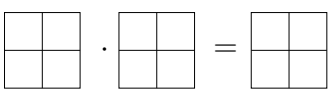
Before we determine the actual structure of the matrix-matrix products, we consider the resulting block structures of multiplying subblock of the $\mathcal{H}$-matrices. Recall section 3.6 of the lecture notes for the following results:

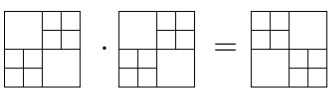
(RR) The product of two $\mathcal{R}$-matrices is an $\mathcal{R}$-matrix ($\mathcal{R} \cdot \mathcal{R} = \mathcal{R}$).

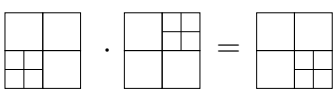
(HR) The product of an $\mathcal{R}$-matrix and an $\mathcal{H}$-matrix is an $\mathcal{R}$-matrix ($\mathcal{R} \cdot \mathcal{H} = \mathcal{R}$, $\mathcal{H} \cdot \mathcal{R} = \mathcal{R}$).

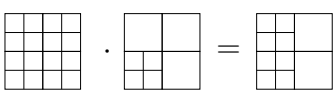
Recall further that the sum of two $\mathcal{R}$ matrices is again an $\mathcal{R}$-matrix. The sum of an $\mathcal{R}$ matrix and an $\mathcal{H}$-matrix is an $\mathcal{H}$-matrix. We now consider the product of different $\mathcal{H}$-matrix blocks and find by using recursion and the above relations:

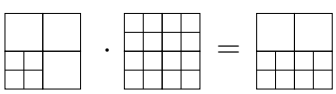
(H1) 

(H2) 

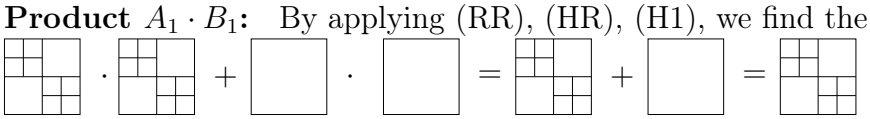
(H3) 

(H4) 

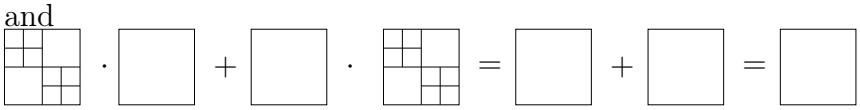
(H5) 

(H6) 

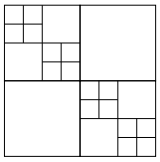
Product $A_1 \cdot B_1$: By applying (RR), (HR), (H1), we find the following intermediate results:



and



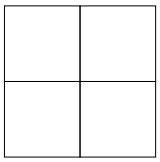
This leads to the following block structure for the matrix-matrix product of A_1 and B_1:

$A_1 \cdot B_1 =$ 

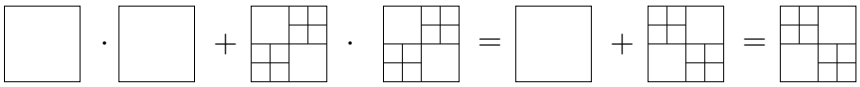
Product $A_2 \cdot B_2$: By applying (RR), (HR), we find the following intermediate results:

and

We find the following (**coarsened**) block structure for the matrix-matrix product of A_2 and B_2:

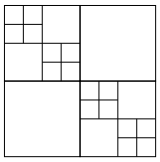
$A_2 \cdot B_2 =$ 

Product $A_3 \cdot B_3$: By applying (RR), (H2), (H3), we find the following intermediate results:

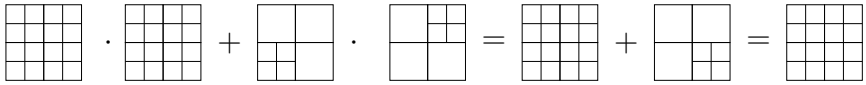


and

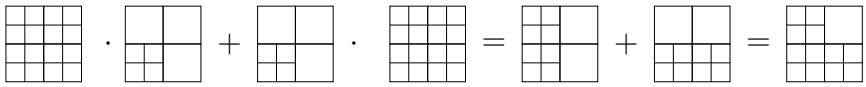
This leads to the following block structure for the matrix-matrix product of A_3 and B_3:

$A_3 \cdot B_3 =$ 

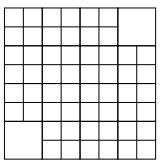
Product $A_4 \cdot B_4$: By applying (RR), (H4), (H5), (H6), we find the following intermediate results:



and



This leads to the following block structure for the matrix-matrix product of A_4 and B_4:

$A_4 \cdot B_4 =$ 

Exercise 4: Let A, B, C be $\mathcal{H}$-matrices with the following 3×3 block structure:

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & 0 \\ A_{31} & 0 & A_{33} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & B_{12} & 0 \\ B_{21} & B_{22} & B_{23} \\ 0 & B_{32} & B_{33} \end{pmatrix}, \quad C = \begin{pmatrix} C_{11} & 0 & C_{13} \\ 0 & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}.$$

Let all the off-diagonal blocks be $\mathcal{R}(k)$-matrices. The diagonal blocks A_{22}, A_{33}, B_{11}, B_{33}, C_{11}, C_{22} have the (recursive) block structure as the matrices A, B, and C. Design algorithms to compute the LU decomposition of these matrices that appropriately exploit the specific block structure (if possible). What do the block structures of the resulting factors of the LU decomposition look like?

Solution hints:

In generel, the block structures of the resulting factors L and U of the LU-decomposition of a matrix with 3×3 block structure look as follows:

$$L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, \quad U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}.$$

For the $\mathcal{H}$-matrices A, B, C we find that the off-diagonal blocks of the resulting factors of the LU-decomposition are $\mathcal{R}(k)$-matrices since the off-diagonal blocks of the $\mathcal{H}$-matrices A, B, C are $\mathcal{R}(k)$-matrices as well.
For the computation of the resulting factors L and U of the LU decomposition of a matrix A, we hence find:

- L_{11} and U_{11} are computed from $A_{11} = L_{11}U_{11}$ where no special block structure can be exploited ($\mathcal{H}$-LU-decomposition without structure).
- U_{12} is computed from $A_{12} = L_{11}U_{12}$ via forward substitution (compare exercise 4 on sheet 04).
- U_{13} is computed from $A_{13} = L_{11}U_{13}$ via forward substitution (compare exercise 4 on sheet 04).
- L_{21} is computed from $A_{21} = L_{21}U_{11}$ with backward substitution.
- L_{31} is computed from $A_{31} = L_{31}U_{11}$ via backward substitution. This step as well as the three previous steps can be computed in parallel.
- L_{22} and U_{22} are computed from $A_{22} - L_{21}U_{12} = L_{22}U_{22}$ where no special block strucure can be exploited ($\mathcal{H}$-LU-decomposition without structure).
- L_{32} is computed from $-L_{31}U_{12} = L_{32}U_{22}$ via backwards substitutino. Here, we use that $A_{32} = 0$ holds.
- U_{23} is computed from $-L_{21}U_{13} = L_{22}U_{23}$ via forward substitution (compare exercise 4 on sheet 04). Here, we exploit that $A_{23} = 0$ holds. This step as well as the previous step can be computed in parallel.

- L_{33} and U_{33} are computed from $A_{33} - L_{31}U_{13} - L_{32}U_{23} = L_{33}U_{33}$ where no special structure can be exploited ($\mathcal{H}$-LU-decomposition without structure).

For the computation of the resulting factors L and U of the LU decomposition of a matrix B, we hence find:

- $U_{13} = 0$ since $0 = B_{13} = L_{11}U_{13}$ holds.
- $L_{31} = 0$ since $0 = B_{31} = L_{31}U_{11}$ holds.
- L_{11} and U_{11} are computed from $B_{11} = L_{11}U_{11}$ where B_{11} has the same block strucure as B ($\mathcal{H}$-LU-decomposition with structure).
- U_{12} is computed from $B_{12} = L_{11}U_{12}$ via forward substitution (compare exercise 4 on sheet 04).
- L_{21} is computed from $B_{21} = L_{21}U_{11}$ via backward substitution. This step and the previous step can be computed in parallel.
- L_{22} and U_{22} are computed from $B_{22} - L_{21}U_{12} = L_{22}U_{22}$ where no special structure can be exploited ($\mathcal{H}$-LU-decomposition without structure).
- U_{23} is computed from $B_{23} = L_{22}U_{23}$ via forward substitution (compare exercise 4 on sheet 04).
- L_{32} is computed from $B_{32} = L_{32}U_{22}$ via backward substitution. This step and the previous step can be computed in parallel.
- L_{33} and U_{33} are computed from $B_{33} - L_{32}U_{23} = L_{33}U_{33}$ where cannot use any special block structure ($\mathcal{H}$-LU-decomposition without structure).

For the computation of the resulting factors L and U of the LU decomposition of a matrix C, we hence find:

- $U_{12} = 0$ since $0 = C_{12} = L_{11}U_{12}$ holds.
- $L_{21} = 0$ since $0 = C_{21} = L_{21}U_{11}$ holds.
- L_{11} and U_{11} are computed from $C_{11} = L_{11}U_{11}$ where C_{11} has the same block structure as C ($\mathcal{H}$-LU-decomposition with structure).
- L_{22} and U_{22} computed from $C_{22} = L_{22}U_{22}$ where C_{22} has the same block structure C ($\mathcal{H}$-LU-decomposition with structure). Here, we use that $L_{21} = 0$ and $U_{12} = 0$. This and the previous step can be computed in parallel.
- U_{13} is computed from $C_{13} = L_{11}U_{13}$ via forward substitution (compare exercise 4 on sheet 04).
- U_{23} is computed from $C_{23} = L_{22}U_{23}$ via forward substitution berechnet (compare exercise 4 on sheet 04).
- L_{31} is computed from $C_{31} = L_{31}U_{11}$ via backward substitution.

- L_{32} is computed from $C_{32} = L_{32}U_{22}$ via backward substitution. This step and the previous three steps can be evaluated in parallel.
- L_{33} and U_{33} are computed from $C_{33} - L_{31}U_{13} - L_{32}U_{23} = L_{33}U_{33}$ where we cannot exploit any special block structure ($\mathcal{H}$-LU-decomposition without structure).

Hint: The structure of the matrix C is hence particularly suitable for the computation of a LU-decomposition. For the matrix A, the calculation of a UL decomposition would instead be advisable.